

Homework #2 - MATH 261A

Noah Arias

Problem 1

We assume that $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$ and are I.I.D.

Part (a)

We know that $\bar{X}_n$ follows a Gaussian distribution, but we must determine the mean and variance parameters.

We do this by a straightforward expectation and variance calculation:

$$E(\bar{X}_n) = E\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{1}{n} \sum_{i=1}^n E(X_i)$$

Since each X_i follows a $N(\mu, \sigma^2)$, its mean is simply μ . Therefore, we can rewrite the last sum in the above calculation as:

$$\frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \frac{\mu \cdot n}{n} = \mu$$

Therefore, we now know that the expectation of $\bar{X}_n$ is μ .

Now we calculate its variance in a similar manner:

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i)$$

We note that the third equality is valid because since the X_i are *independent* as well as identically distributed. This means that any and all covariance terms are precisely zero. Continuing the calculation, we arrive at:

$$\frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2 \cdot n}{n^2} = \frac{\sigma^2}{n}$$

With this result, we can finally conclude that $\bar{X}_n \sim N(\mu, \frac{\sigma^2}{n})$.

Part (b)

The general structure of a confidence interval is following:

$$(\text{point estimate} \pm \text{critical value} \cdot \text{standard deviation/error})$$

We are specifically deriving a 95% confidence interval for the sampling distribution of $\bar{X}_n$ which we now know is $N(\mu, \frac{\sigma^2}{n})$.

The first thing we recognize is that μ is the parameter, whose location we are trying to ascertain with the estimate $\bar{X}_n$, for a given sample of size n . Thus, the point estimate for μ is $\bar{X}_n$.

Secondly, we must determine the critical value. This is dependent on the sampling distribution which is $N(\mu, \frac{\sigma^2}{n})$. Since we know not only our distribution is Normal but the population variance too, our critical value is a z-score with form:

$$z_{\text{crit}} = \frac{\bar{x}_n - \mu}{\sigma/\sqrt{n}}$$

This expression follows a standard Normal distribution $N(0, 1)$.

The standard error is easy to find as it is simply $\frac{\sigma}{\sqrt{n}}$.

Starting at the requirement that for a 95% confidence interval, the following must be true:

$$P(-1.96 < \frac{\bar{x}_n - \mu}{\sigma/\sqrt{n}} < 1.96) = 0.95$$

To arrive at our desired confidence interval, we make take the inequality inside the probability operation and solve for μ :

$$\begin{aligned}
-1.96 &< \frac{\bar{x}_n - \mu}{\sigma/\sqrt{n}} < 1.96 \\
-1.96 \cdot \frac{\sigma}{\sqrt{n}} &< \bar{x}_n - \mu < 1.96 \cdot \frac{\sigma}{\sqrt{n}} \\
-\bar{x}_n - 1.96 \cdot \frac{\sigma}{\sqrt{n}} &< -\mu < -\bar{x}_n + 1.96 \cdot \frac{\sigma}{\sqrt{n}} \\
\bar{x}_n - 1.96 \cdot \frac{\sigma}{\sqrt{n}} &< \mu < \bar{x}_n + 1.96 \cdot \frac{\sigma}{\sqrt{n}}
\end{aligned}$$

Therefore:

$$P\left(\bar{X}_n - 1.96 \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{x}_n + 1.96 \cdot \frac{\sigma}{\sqrt{n}}\right) = 0.95$$

Part (c)

We now assume that there is another I.I.D. sample $X_{n+1}, \dots, X_{2n}$ of size n , that is also independent of the first sample, and is from another Normal distribution $N(\mu, \tau^2)$. (We also assume that τ^2 does **NOT** refer to the precision, but instead the variance of the second sample.)

Immediately, we can conclude that the sampling distribution of $\bar{X}_{2n}$ is yet another Normal distribution.